Analyses

TerraLID ID

ID and name: A0 terralid_analysis_id
Provided by: TerraLID system
Obligation: mandatory
Occurrences: 1
Definition: The ID of the analysis in the TerraLID database.
Allowed values and other constraints: t.b.d.
Example: t.b.d.

1. Laboratory ID

ID and name: A1 analysis_lab_id
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: The ID(s) of the analysis in a laboratory and/or another database.
Allowed values and other constraints: free text
Example: 2024-TR01

2. Analysis type

ID and name: A2 analysis_lia_type
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The type of analysis for measuring the lead isotope composition.
Allowed values and other constraints: controlled vocabulary

3. Preparation protocol

ID and name: A3 analysis_lia_preparation
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: Information about how the sample was prepared for analysis.

with the two subproperties:

3.1 Description

ID and name: A3.1 analysis_lia_preparation_description
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: The way the sample was prepared for analysis.
Allowed values and other constraints: free text
Example: Dissolution with aqua regia at 80°C and evaporation to dryness, followed by dissolution in 2% HNO_3.

3.2 Publication

ID and name: A3.2 analysis_lia_preparation_publication
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: If the preparation protocol was already published, the publication of the protocol.

with the five subproperties:

3.2.1 Persistent Identifier

ID and name: B5.1 relation_pid
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: The persistent identifier or TerraLID ID associated with a resource or related research output. Mandatory if referring to another entity in the TerraLID database; in this case, the TerraLID ID must be used and `B5.1.2 Type` is "TerraLID". Within a submission, the identifiers of the submitted records may be used; the TerraLID system replaces them with the assigned TerraLID IDs.

with the two subproperties:

3.2.1.1 VALUE

ID and name: B5.1.1 relation_pid_value
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: Value of the persistent identifier or TerraLID ID. **Allowed values and other constraints:** valid persistent identifier according to the options listed in `B5.1.2 Type`, or a TerraLID ID if `B5.1.2 Type` = "TerraLID". **Example:** 10.60510/ICDP5054ESYI201

3.2.1.2 TYPE

ID and name: B5.1.2 relation_pid_type
Provided by: data provider
Obligation: mandatory

Occurrences: 1
Definition: The type of the identifier.
Allowed values and other constraints: controlled vocabulary (identifier). Use "TerraLID" for entities in the TerraLID database.

3.2.2 Full reference

ID and name: B5.2 relation_text
Provided by: data provider, API (DOI)
Obligation: recommended
Occurrences: 0–1
Definition: The full reference of a publication.
Allowed values and other constraints: free text
Example: Palinkaš, L. A., 1985, Lead isotope patterns in galenas from some selected ore deposits in Croatia and NW Bosnia, Geološki Vjesnik, 38, 175–89.

3.2.3 Kind of relation

ID and name: B5.3 relation_kind
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: Relationship between the item and the related resource, research output, or TerraLID entity (e.g., IsPartOf for a sample taken from an object). **Allowed values and other constraints:** controlled vocabulary

3.2.4 Type of resource

ID and name: B5.4 relation_resource
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: Type of resource or research output. Mandatory unless `B5.1.2 Type` = "TerraLID", as the type of a TerraLID entity follows from its TerraLID ID. **Allowed values and other constraints:** controlled vocabulary

3.2.5 Additional details

ID and name: B5.5 relation_detail
Provided by: data provider
Obligation: optional
Occurrences: 0–n
Definition: Additional information about the relation, e.g., if the related work addresses a specific aspect of the item (e.g., the geological setting).
Allowed values and other constraints: free text
Example: Information about the dating of the site.

4. Analysed material

ID and name: A4 analysis_lia_material
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: Capture here if specific materials within a sample are analysed, such as mineral separates or laser ablation of individual mineral species within the same specimen.
Allowed values and other constraints: free text
Example: only malachite analysed

5. Separation protocol

ID and name: A5 analysis_lia_separation
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: Information about the protocol used for the separation of lead from the sample matrix.

with the two subproperties:

5.1 Description

ID and name: A5.1 analysis_lia_separation_description
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: If unpublished, provide description of the protocol used for the separation of lead from the sample matrix. Otherwise, list all deviations from the published protocol.
Allowed values and other constraints: free text
Example: t.b.d.

5.2 Publication

ID and name: A5.2 analysis_lia_separation_publication
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: The publication of the protocol used for separating lead from the sample matrix.

with the five subproperties:

5.2.1 Persistent Identifier

ID and name: B5.1 relation_pid
Provided by: data provider
Obligation: recommended
Occurrences: 0–n

Definition: The persistent identifier or TerraLID ID associated with a resource or related research output. Mandatory if referring to another entity in the TerraLID database; in this case, the TerraLID ID must be used and `B5.1.2 Type` is "TerraLID". Within a submission, the identifiers of the submitted records may be used; the TerraLID system replaces them with the assigned TerraLID IDs.

with the two subproperties:

5.2.1.1 VALUE

ID and name: B5.1.1 relation_pid_value
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: Value of the persistent identifier or TerraLID ID. **Allowed values and other constraints:** valid persistent identifier according to the options listed in `B5.1.2 Type`, or a TerraLID ID if `B5.1.2 Type` = "TerraLID". **Example:** 10.60510/ICDP5054ESYI201

5.2.1.2 TYPE

ID and name: B5.1.2 relation_pid_type
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The type of the identifier.
Allowed values and other constraints: controlled vocabulary (identifier). Use "TerraLID" for entities in the TerraLID database.

5.2.2 Full reference

ID and name: B5.2 relation_text
Provided by: data provider, API (DOI)
Obligation: recommended
Occurrences: 0–1
Definition: The full reference of a publication.
Allowed values and other constraints: free text
Example: Palinkaš, L. A., 1985, Lead isotope patterns in galenas from some selected ore deposits in Croatia and NW Bosnia, Geološki Vjesnik, 38, 175–89.

5.2.3 Kind of relation

ID and name: B5.3 relation_kind
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: Relationship between the item and the related resource, research output, or TerraLID entity (e.g., IsPartOf for a sample taken from an object). **Allowed values and other constraints:** controlled vocabulary

5.2.4 Type of resource

ID and name: B5.4 relation_resource
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: Type of resource or research output. Mandatory unless `B5.1.2 Type` = "TerraLID", as the type of a TerraLID entity follows from its TerraLID ID. **Allowed values and other constraints:** controlled vocabulary

5.2.5 Additional details

ID and name: B5.5 relation_detail
Provided by: data provider
Obligation: optional
Occurrences: 0–n
Definition: Additional information about the relation, e.g., if the related work addresses a specific aspect of the item (e.g., the geological setting).
Allowed values and other constraints: free text
Example: Information about the dating of the site.

6. Measurement device

ID and name: A6 analysis_lia_instrument
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: Information about the instrument used to measure the lead isotope composition.

with the three subproperties:

6.1 Instrument type

ID and name: A6.1 analysis_lia_instrument_type
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The type of instrument.
Allowed values and other constraints: controlled vocabulary

6.2 Instrument model

ID and name: A6.2 analysis_lia_instrument_model
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: The manufacturer and model name of the instrument.
Allowed values and other constraints: controlled vocabulary

6.3 Persistent identifier (PIDinst)

ID and name: A6.3 analysis_lia_instrument_pid
Provided by: data provider
Obligation: optional
Occurrences: 0–1
Definition: The full URL of the instrument's PIDinst.
Allowed values and other constraints: is valid PIDinst
Example: http://hdl.handle.net/21.11157/cd5777a9-07c4-4e80-a770-9f294f09894d

7. Analyte Pb concentration

ID and name: A7 analysis_lia_pb_concentration
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: The lead concentration of the analyte.

with the eight subproperties:

7.1 Analytical method

ID and name: B4.1 chemistry_method
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The method used to determine the chemical composition.
Allowed values and other constraints: controlled vocabulary

7.2 Analysed compound

ID and name: B4.2 chemistry_compound
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: The analysed chemical compound (chemical element, oxide, isotope).
Allowed values and other constraints: controlled vocabulary, the actual measured compound must be reported if available (e.g., the isotope for mass spectrometric values rather than the element).

7.3 Value

ID and name: B4.3 chemistry_value
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: The concentration of the analysed chemical compound.

with the two subproperties:

7.3.1 ABSOLUTE VALUE

ID and name: B4.3.1 chemistry_value_absolute
Provided by: data provider
Obligation: recommended
Occurrences: 1–n
Definition: The concentration of the analysed chemical compound.
Allowed values and other constraints: decimal number
Example: 15.3

7.3.2 QUALITATIVE VALUE

ID and name: B4.3.2 chemistry_value_relative
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: Value of qualitative results and other non-numeric values such as the reporting of detection or quantification limits rather than values.
Allowed values and other constraints: free text
Example: +++, tr, <0.4, bdl = 0.5 wt%

7.4 Unit

ID and name: B4.4 chemistry_unit
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: The unit in which the concentration of the analysed chemical compound is given.
Allowed values and other constraints: controlled vocabulary

7.5 Uncertainty type

ID and name: B4.5 chemistry_uncertainty_type
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: The type of analytical uncertainty.
Allowed values and other constraints: controlled vocabulary

7.6 Confidence level

ID and name: B4.6 chemistry_uncertainty_sigma
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: Multiple of standard deviation (sigma) of the reported absolute analytical uncertainty.

Allowed values and other constraints: 1, 2, 3
Example: 2

7.7 Uncertainty value

ID and name: B4.7 chemistry_uncertainty_value
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: Value of the absolute analytical uncertainty.
Allowed values and other constraints: decimal number
Example: 0.3

8. Mean total intensity of analyte

ID and name: A8 analysis_lia_pb_intensity
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: The mean total intensity (sum of all isotope signals) during the analysis.

with the two subproperties:

8.1 Value

ID and name: A8.1 analysis_lia_pb_intensity_value
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The value of the mean total intensity.
Allowed values and other constraints: decimal number
Example: 40.5

8.2 Unit

ID and name: A8.2 analysis_lia_pb_intensity_unit
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The SI unit in which the mean total intensity is given.
Allowed values and other constraints: controlled vocabulary

9. Reference materials

ID and name: A9 analysis_lia_standard-pb
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: Information about the reference material(s) for lead isotopes used during the analysis for quality control and/or correction of instrumental fractionation.

with the six subproperties:

9.1 Name of lead isotope reference material

ID and name: A9.1 analysis_lia_standard-pb_name
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: The name of the reference material for lead isotopes.
Allowed values and other constraints: controlled vocabulary

9.2 Publication of lead isotope reference material

ID and name: A9.2 analysis_lia_standard-pb_publication
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: Publication reporting the values of the reference material's lead isotope ratios used for mass bias correction. These are not necessarily the values originally published for the reference material.

with the five subproperties:

9.2.1 Persistent Identifier

ID and name: B5.1 relation_pid
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: The persistent identifier or TerraLID ID associated with a resource or related research output. Mandatory if referring to another entity in the TerraLID database; in this case, the TerraLID ID must be used and `B5.1.2 Type` is "TerraLID". Within a submission, the identifiers of the submitted records may be used; the TerraLID system replaces them with the assigned TerraLID IDs.

with the two subproperties:

9.2.1.1 VALUE

ID and name: B5.1.1 relation_pid_value
Provided by: data provider
Obligation: mandatory
Occurrences: 1

Definition: Value of the persistent identifier or TerraLID ID. **Allowed values and other constraints:** valid persistent identifier according to the options listed in `B5.1.2 Type`, or a TerraLID ID if `B5.1.2 Type` = "TerraLID". **Example:** 10.60510/ICDP5054ESYI201

9.2.1.2 TYPE

ID and name: B5.1.2 relation_pid_type
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The type of the identifier.
Allowed values and other constraints: controlled vocabulary (identifier). Use "TerraLID" for entities in the TerraLID database.

9.2.2 Full reference

ID and name: B5.2 relation_text
Provided by: data provider, API (DOI)
Obligation: recommended
Occurrences: 0–1
Definition: The full reference of a publication.
Allowed values and other constraints: free text
Example: Palinkaš, L. A., 1985, Lead isotope patterns in galenas from some selected ore deposits in Croatia and NW Bosnia, Geološki Vjesnik, 38, 175–89.

9.2.3 Kind of relation

ID and name: B5.3 relation_kind
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: Relationship between the item and the related resource, research output, or TerraLID entity (e.g., IsPartOf for a sample taken from an object). **Allowed values and other constraints:** controlled vocabulary

9.2.4 Type of resource

ID and name: B5.4 relation_resource
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: Type of resource or research output. Mandatory unless `B5.1.2 Type` = "TerraLID", as the type of a TerraLID entity follows from its TerraLID ID. **Allowed values and other constraints:** controlled vocabulary

9.2.5 Additional details

ID and name: B5.5 relation_detail
Provided by: data provider
Obligation: optional
Occurrences: 0–n

Definition: Additional information about the relation, e.g., if the related work addresses a specific aspect of the item (e.g., the geological setting).
Allowed values and other constraints: free text
Example: Information about the dating of the site.

9.3 Measured values of lead isotope reference material

ID and name: A9.3 analysis_lia_standard-pb_measured
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: The measured lead isotope ratios of the reference material.

with the seven subproperties:

9.3.1 Name

ID and name: B6.1 lia_ratio_name
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The lead isotope ratio for which the value is reported.
Allowed values and other constraints: 206Pb/204Pb, 207Pb/204Pb, 208Pb/204Pb, 204Pb/206Pb, 207Pb/206Pb, 208Pb/206Pb, 207Pb/208Pb, 206Pb/208Pb
Example: 206Pb/204Pb

9.3.2 Value

ID and name: B6.2 lia_ratio_value
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: Value of the lead isotope ratio.
Allowed values and other constraints: decimal number
Example: 18.59123

9.3.3 Uncertainty

ID and name: B6.3 lia_ratio_uncertainty_type
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: Type of analytical uncertainty for the lead isotope ratio.
Allowed values and other constraints: controlled vocabulary

9.3.4 Confidence level

ID and name: B6.4 lia_ratio_uncertainty_sigma
Provided by: data provider

Obligation: recommended
Occurrences: 0–1
Definition: Multiple of analytical uncertainty (sigma) of the reported absolute analytical uncertainty.
Allowed values and other constraints: 1, 2, 3
Example: 2

9.3.5 Absolute uncertainty

ID and name: B6.5 lia_ratio_uncertainty_value_absolute
Provided by: data provider, TerraLID system
Obligation: recommended
Occurrences: 0–1
Definition: Value of the reported absolute analytical uncertainty.
Allowed values and other constraints: decimal number
Example: 0.00008

9.3.6 Relative uncertainty

ID and name: B6.6 lia_ratio_uncertainty_value_relative
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: Value of relative analytical uncertainty for the lead isotope ratio in per cent (%). If provided, the TerraLID system will calculate the corresponding absolute values.
Allowed values and other constraints: decimal number
Example: 0.1

9.3.7 Source

ID and name: B6.7 lia_ratio_source
Provided by: TerraLID system
Obligation: mandatory
Occurrences: 1
Definition: Whether the date was reported in the publication or calculated by the TerraLID system from other published values.
Allowed values and other constraints: original, calculated
Example: original

9.4 Name of thallium isotope reference material

ID and name: A9.4 analysis_lia_standard-tl_name
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: The name of the thallium reference material used during the analysis for quality control and/or correction of instrumental fractionation.
Allowed values and other constraints: controlled vocabulary

9.5 Measured $^{205}Tl/^{203}Tl$ ratio of thallium isotope reference material

ID and name: A9.5 analysis_lia_standard-tl_measured
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: The measured $^{205}Tl/^{203}Tl$ ratio of the reference material.
Allowed values and other constraints: decimal number
Example: 2.38714

9.6 Concentration of the thallium isotope reference material

ID and name: A9.6 analysis_lia_standard-tl_concentration
Provided by: data provider
Obligation: optional
Occurrences: 0–1
Definition: The concentration of the thallium reference material added to the sample in ppb (e.g. ng/g, µg/l).
Allowed values and other constraints: number
Example: 100

10. Mass bias correction model

ID and name: A10 analysis_lia_correction
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: The model(s) used for mass bias correction of the lead isotope data.
Allowed values and other constraints: controlled vocabulary

11. Laboratory

ID and name: A11 analysis_lia_laboratory
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: The laboratory that performed the lead isotope analysis.

with the nine subproperties:

11.1 Role

ID and name: B1.1 person_role
Provided by: data provider
Obligation: mandatory

Occurrences: 1–n
Definition: The role in which the person is related to the linked information.
Allowed values and other constraints: controlled vocabulary
Example: Author

11.2 First names

ID and name: B1.2 person_name_first
Provided by: data provider, API (ORCID ID)
Obligation: recommended
Occurrences: 0–1
Definition: The first and middle name(s) of the person.
Allowed values and other constraints: free text
Example: Jane

11.3 Last Name

ID and name: B1.3 person_name_last
Provided by: data provider, API (ORCID ID)
Obligation: mandatory
Occurrences: 1
Definition: The last name of the person.
Allowed values and other constraints: free text
Example: Doe

11.4 Persistent Identifier

ID and name: B1.4 person_pid
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: Persistent identifier(s) assigned to the person.

with the two subproperties:

11.4.1 VALUE OF PERSISTENT IDENTIFIER

ID and name: B1.4.1 person_pid_value
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: A persistent identifier assigned to the analysed material.
Allowed values and other constraints: Valid persistent identifier according to B1.4.2 Type of persistent identifier
Example: 0000–0001–2345–678X

11.4.2 TYPE OF PERSISTENT IDENTIFIER

ID and name: B1.4.2 person_pid_type
Provided by: data provider
Obligation: mandatory

Occurrences: 1
Definition: The type of the persistent identifier.
Allowed values and other constraints: controlled vocabulary

11.5 Affiliation name

ID and name: B1.5 person_affiliation_name
Provided by: data provider, API (ORCID ID, ROR ID)
Obligation: mandatory
Occurrences: 1–n
Definition: The name of the person's affiliation.
Allowed values and other constraints: free text
Example: Institute of Time Travels

11.6 ROR ID

ID and name: B1.6 person_affiliation_ror
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: The ROR of the person's affiliation.
Allowed values and other constraints: is valid ROR ID
Example: 09af7gtg53

11.7 Address

ID and name: B1.7 person_affiliation_address
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: The address of the person's affiliation.
Allowed values and other constraints: free text
Example: Teleporter Avenue 123, Ankh–Morpok, United States of Humanities

11.8 Mail address

ID and name: B1.8 person_mail
Provided by: data provider, API (ORCID ID)
Obligation: recommended
Occurrences: 0–n
Definition: The mail address of the person.
Allowed values and other constraints: is valid mail address
Example: jane.doe@timetravels.int

11.9 Website

ID and name: B1.9 person_url
Provided by: data provider, API (ORCID ID)
Obligation: optional

Occurrences: 0–1
Definition: The URL of a person.
Allowed values and other constraints: is valid URL
Example: https://www.timetravels.int/members/jane–doe

12. Date of analysis

ID and name: A12 analysis_lia_date
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: The day of the analysis.
Allowed values and other constraints: date formatted as YYYY-MM-DD
Example: 2024-02-24

13. Description

ID and name: A13 analysis_lia_description
Provided by: data provider
Obligation: optional
Occurrences: 0–1
Definition: Additional information about the analytical procedure not captured elsewhere.
Allowed values and other constraints: free text
Example: t.b.d.

14. Lead isotope ratios

ID and name: A14 analysis_lia_ratio
Provided by: data provider, TerraLID system
Obligation: mandatory
Occurrences: 1–n
Definition: Mass-bias corrected lead isotope ratios and analytical uncertainty. The TerraLID system will calculate all ratios not reported in the original publication.

with the seven subproperties:

14.1 Name

ID and name: B6.1 lia_ratio_name
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The lead isotope ratio for which the value is reported.
Allowed values and other constraints: 206Pb/204Pb, 207Pb/204Pb, 208Pb/204Pb, 204Pb/206Pb, 207Pb/206Pb,

208Pb/206Pb, 207Pb/208Pb, 206Pb/208Pb
Example: 206Pb/204Pb

14.2 Value

ID and name: B6.2 lia_ratio_value
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: Value of the lead isotope ratio.
Allowed values and other constraints: decimal number
Example: 18.59123

14.3 Uncertainty

ID and name: B6.3 lia_ratio_uncertainty_type
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: Type of analytical uncertainty for the lead isotope ratio.
Allowed values and other constraints: controlled vocabulary

14.4 Confidence level

ID and name: B6.4 lia_ratio_uncertainty_sigma
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: Multiple of analytical uncertainty (sigma) of the reported absolute analytical uncertainty.
Allowed values and other constraints: 1, 2, 3
Example: 2

14.5 Absolute uncertainty

ID and name: B6.5 lia_ratio_uncertainty_value_absolute
Provided by: data provider, TerraLID system
Obligation: recommended
Occurrences: 0–1
Definition: Value of the reported absolute analytical uncertainty.
Allowed values and other constraints: decimal number
Example: 0.00008

14.6 Relative uncertainty

ID and name: B6.6 lia_ratio_uncertainty_value_relative
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: Value of relative analytical uncertainty for the lead isotope ratio in per cent (%). If provided, the TerraLID

system will calculate the corresponding absolute values.
Allowed values and other constraints: decimal number
Example: 0.1

14.7 Source

ID and name: B6.7 lia_ratio_source
Provided by: TerraLID system
Obligation: mandatory
Occurrences: 1
Definition: Whether the date was reported in the publication or calculated by the TerraLID system from other published values.
Allowed values and other constraints: original, calculated
Example: original

15. Age model parameters

ID and name: A15 analysis_lia_age_model
Provided by: TerraLID system
Obligation: recommended
Occurrences: 0–n
Definition: Age model parameters calculated from the mass-bias corrected lead isotope ratios.

with the nine subproperties:

15.1 Age model name

ID and name: A15.1 analysis_lia_age_model_name
Provided by: TerraLID system
Obligation: mandatory
Occurrences: 1
Definition: The age model used for calculating the parameters
Allowed values and other constraints: SK75, CR75, AJ84, representing the age models defined by Stacey & Kramers (1975), Cumming & Richards (1975), and Albarède & Juteau (1984), respectively.
Example: SK75

15.2 Model age

ID and name: A15.2 analysis_lia_age_model_Tmod
Provided by: TerraLID system
Obligation: recommended
Occurrences: 0–1
Definition: Value of the model age in million years (Ma).
Allowed values and other constraints: decimal number
Example: 250.54

15.3 Uncertainty of model age

ID and name: A15.3 analysis_lia_age_model_Tmod_uncertainty
Provided by: TerraLID system
Obligation: recommended
Occurrences: 0–1
Definition: Uncertainty of the model age.
Allowed values and other constraints: decimal number
Example: t.b.d.

15.4 Mu

ID and name: A15.4 analysis_lia_age_model_mu
Provided by: TerraLID system
Obligation: recommended
Occurrences: 0–1
Definition: Value of mu (μ).
Allowed values and other constraints: decimal number
Example: 9.86

15.5 Uncertainty of mu

ID and name: A15.5 analysis_lia_age_model_mu_uncertainty
Provided by: TerraLID system
Obligation: recommended
Occurrences: 0–1
Definition: Uncertainty of mu.
Allowed values and other constraints: decimal number
Example: t.b.d.

15.6 Kappa

ID and name: A15.6 analysis_lia_age_model_kappa
Provided by: TerraLID system
Obligation: recommended
Occurrences: 0–1
Definition: Value of kappa (κ).
Allowed values and other constraints: decimal number
Example: 3.92

15.7 Uncertainty of kappa

ID and name: A15.7 analysis_lia_age_model_kappa_uncertainty
Provided by: TerraLID system

Obligation: recommended
Occurrences: 0–1
Definition: Uncertainty of kappa.
Allowed values and other constraints: decimal number
Example: t.b.d.

15.8 Omega

ID and name: A15.8 analysis_lia_age_model_omega
Provided by: TerraLID system
Obligation: recommended
Occurrences: 0–1
Definition: Value of omega (ω).
Allowed values and other constraints: decimal number
Example: t.b.d.

15.9 Uncertainty of omega

ID and name: A15.9 analysis_lia_age_model_omega_uncertainty
Provided by: TerraLID system
Obligation: recommended
Occurrences: 0–1
Definition: Uncertainty of omega.
Allowed values and other constraints: decimal number
Example: t.b.d.

16. Relations

ID and name: A16 analysis_lia_relation
Provided by: data provider
Obligation: recommended
Occurrences: –n
Definition: Information about related entities and other research output such as publications providing relevant information about the analysis. This includes any literature from which information about the analysis was extracted.

with the five subproperties:

16.1 Persistent Identifier

ID and name: B5.1 relation_pid
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: The persistent identifier or TerraLID ID associated with a resource or related research output. Mandatory if referring to another entity in the TerraLID database; in this case, the TerraLID ID must be used and `B5.1.2 Type` is

"TerraLID". Within a submission, the identifiers of the submitted records may be used; the TerraLID system replaces them with the assigned TerraLID IDs.

with the two subproperties:

16.1.1 VALUE

ID and name: B5.1.1 relation_pid_value
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: Value of the persistent identifier or TerraLID ID. **Allowed values and other constraints:** valid persistent identifier according to the options listed in `B5.1.2 Type`, or a TerraLID ID if `B5.1.2 Type` = "TerraLID". **Example:** 10.60510/ICDP5054ESYI201

16.1.2 TYPE

ID and name: B5.1.2 relation_pid_type
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The type of the identifier.
Allowed values and other constraints: controlled vocabulary (identifier). Use "TerraLID" for entities in the TerraLID database.

16.2 Full reference

ID and name: B5.2 relation_text
Provided by: data provider, API (DOI)
Obligation: recommended
Occurrences: 0–1
Definition: The full reference of a publication.
Allowed values and other constraints: free text
Example: Palinkaš, L. A., 1985, Lead isotope patterns in galenas from some selected ore deposits in Croatia and NW Bosnia, Geološki Vjesnik, 38, 175–89.

16.3 Kind of relation

ID and name: B5.3 relation_kind
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: Relationship between the item and the related resource, research output, or TerraLID entity (e.g., IsPartOf for a sample taken from an object). **Allowed values and other constraints:** controlled vocabulary

16.4 Type of resource

ID and name: B5.4 relation_resource
Provided by: data provider
Obligation: mandatory

Occurrences: 1–n
Definition: Type of resource or research output. Mandatory unless `B5.1.2 Type` = "TerraLID", as the type of a TerraLID entity follows from its TerraLID ID. **Allowed values and other constraints:** controlled vocabulary

16.5 Additional details

ID and name: B5.5 relation_detail
Provided by: data provider
Obligation: optional
Occurrences: 0–n
Definition: Additional information about the relation, e.g., if the related work addresses a specific aspect of the item (e.g., the geological setting).
Allowed values and other constraints: free text
Example: Information about the dating of the site.